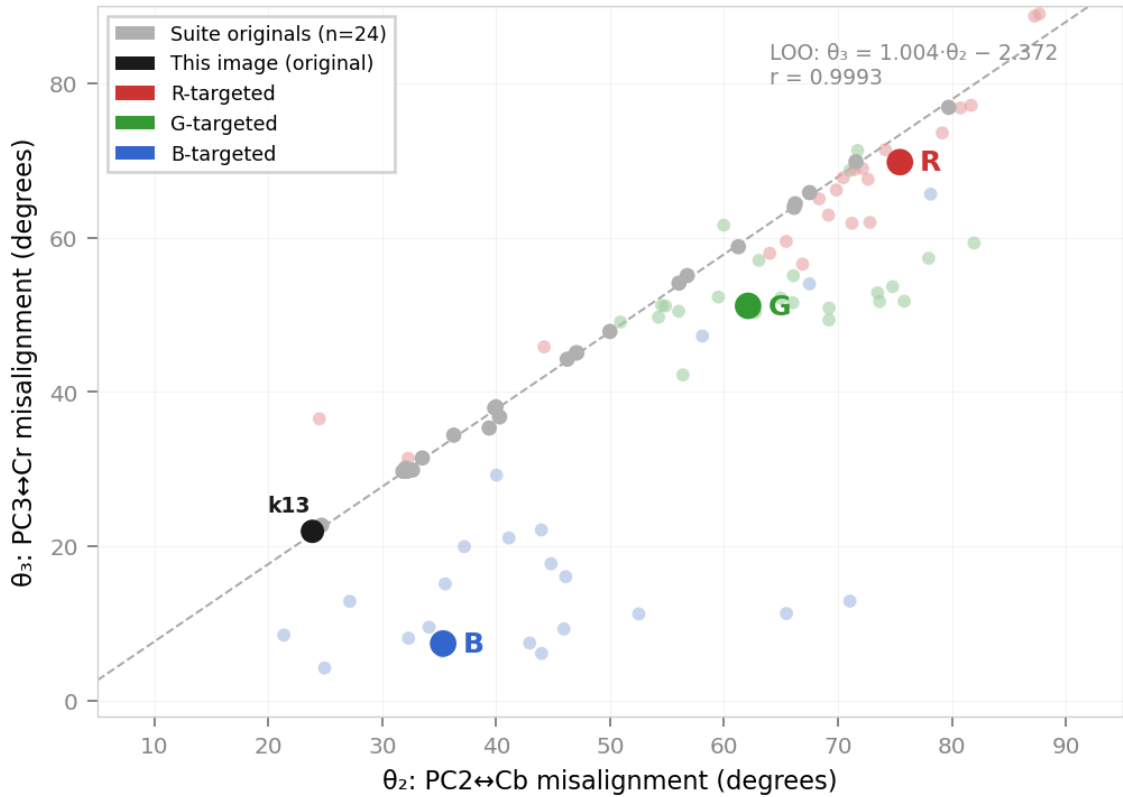


Compression Profile: KODIM13768x512 | 24-bit RGB | PNG (lossless)

mountain stream | Near-Degenerate (PC1 = 97.6%)

1. Geometric Displacement Under Directional Perturbation



2. Pairwise Correlation and Angular Misalignment

Version	R-G	R-B	G-B	Avg r	theta2	theta3	L00 dev
Original	0.9818	0.9469	0.9639	0.964	23.86°	21.94°	+0.36°
R-targeted	0.2190	0.2361	0.9336	0.463	75.44°	69.82°	-3.55°
G-targeted	0.2533	0.9359	0.2563	0.482	62.11°	51.18°	-8.81°
B-targeted	0.9510	0.0580	0.0561	0.355	35.34°	7.39°	-25.72°

R-targeted isolates red (R-G, R-B < 0.24) while preserving G-B coupling at 0.934.
G-targeted isolates green (R-G, G-B < 0.26) while preserving R-B at 0.936.
B-targeted preserves R-G at 0.951 while driving R-B and G-B near zero.
All three perturbation directions depart from the L00 constraint manifold (r = 0.9993).

Compression Profile: KODIM13

RGB | PNG (lossless)

mountain stream | Near-Degenerate (PC1 = 97.6%)

768x512 | 24-bit

3. File Format and Size Comparison

All versions depict the same scene. BPP = (file size in bits) / (pixel count).

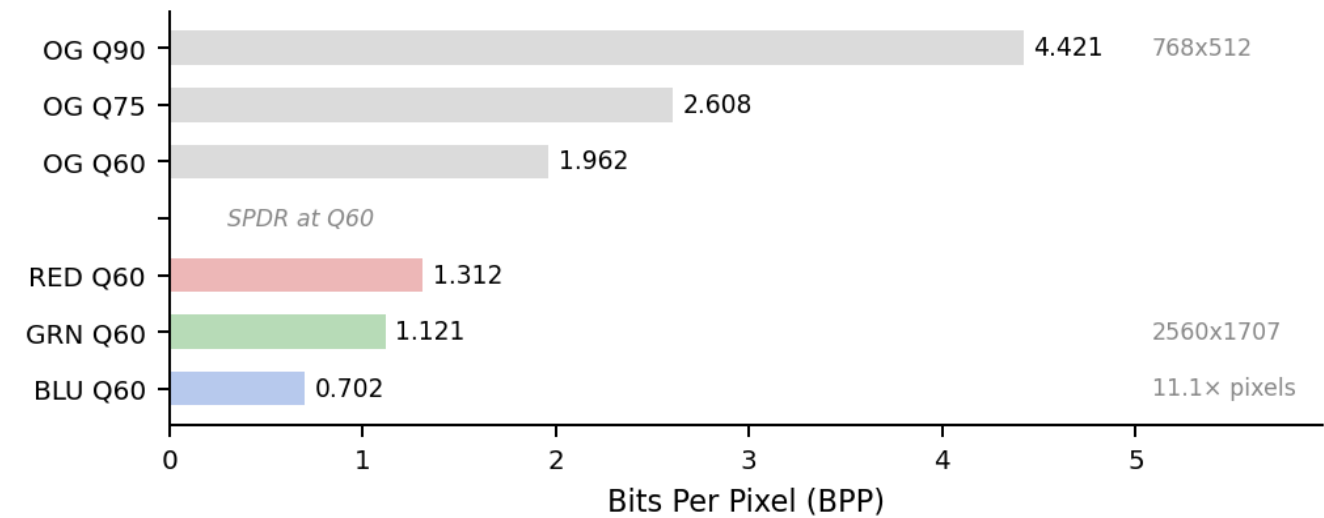
ORIGINAL (kodim13)

Version	Format	Resolution	Size	BPP	Pixels
PNG (lossless)	PNG	768x512	803.4 KB	16.738	393,216
JPEG Q90	JPEG	768x512	212.2 KB	4.421	393,216
JPEG Q75	JPEG	768x512	125.2 KB	2.608	393,216
JPEG Q60	JPEG	768x512	94.2 KB	1.962	393,216

REDISTRIBUTED (kodim13 SPDR at Q60)

Version	Format	Resolution	Size	BPP	Pixels
RED Q60	JPEG	2560x1707	699.6 KB	1.312	4,369,920
GRN Q60	JPEG	2560x1707	598.0 KB	1.121	4,369,920
BLU Q60	JPEG	2560x1707	374.5 KB	0.702	4,369,920

4. Quality Ladder: SPDR Q60 vs Original Q90



All three perturbation directions at Q60 produced lower BPP than the original at Q90, while carrying 11.1× the original pixel count.

Compression Profile: KODIM13

RGB | PNG (lossless)

mountain stream | Near-Degenerate (PC1 = 97.6%)

768x512 | 24-bit

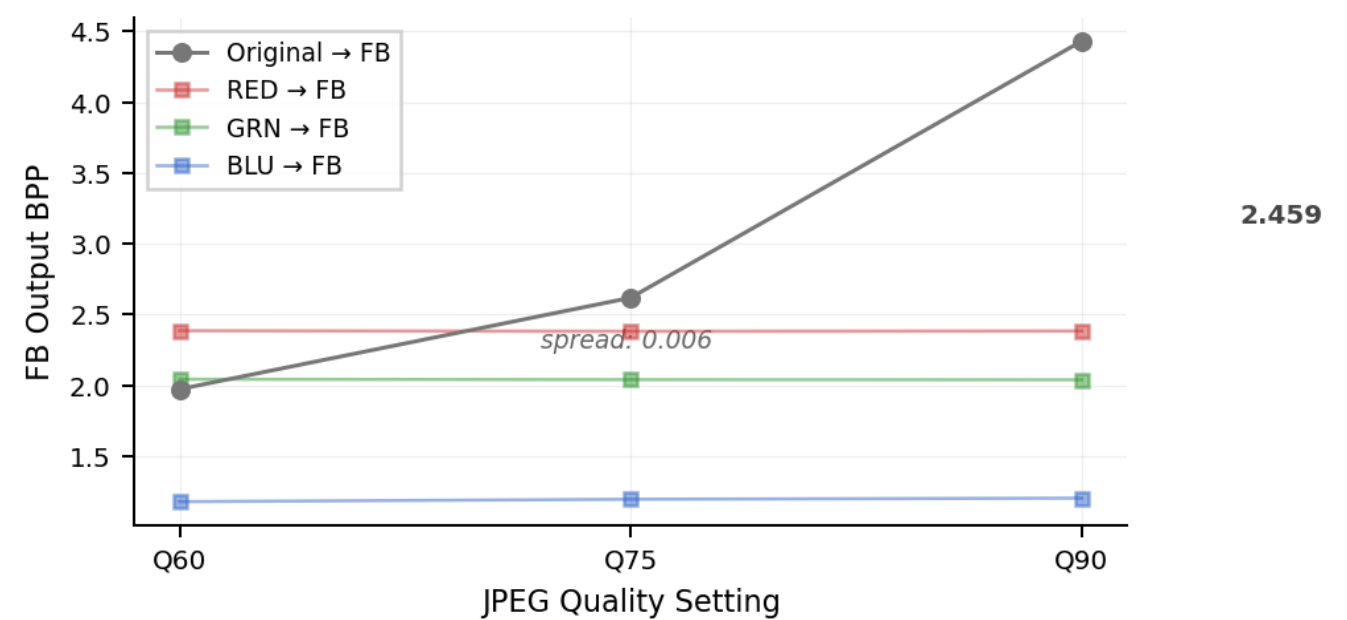
5. Third-Party Compression Pipeline (Facebook/Meta)

Each version uploaded to Facebook. Platform-compressed output measured at steady state (FB2).

Pipeline	Input Res	Output Res	FB BPP	BPP reduction
OG PNG → FB	768x512	768x512	4.192	—
R-targeted → FB	2560x1707	2048x1366	2.383	43.2%
G-targeted → FB	2560x1707	2048x1366	2.038	51.4%
B-targeted → FB	2560x1707	2048x1366	1.202	71.3%
ABC (no perturbation) → FB	2560x1707	2048x1366	1.599	61.9%

ABC control: same pipeline, same resolution, no covariance perturbation. Avg |r| unchanged (0.9638 vs 0.964 original). Comparable BPP reduction was not observed in the ABC control.

6. Quality Parameter Sensitivity Through Facebook



Original: Q60 → Q90 produces a 2.459 BPP spread through Facebook.
Redistributed: Q60 → Q90 produces a 0.006 BPP spread through Facebook.
Redistributed versions produced substantially smaller Q60-90 BPP variation through the measured pipeline.

7. Pipeline Summary – kodim13

Best BPP (Q60, direct)	0.702	2560x1707, 4,369,920 px
Best BPP through FB	1.202	2048x1366 output
vs OG Q90 → FB	72.9%	SPDR Q60→FB vs OG Q90→FB
Pixel ratio	11.1×	4,369,920 vs 393,216